

Notes: Backus, Chernov, Zin, and Zviadadze (RFS, 2022)

Monetary Policy Risk: Rules versus Discretion

Contents

1	Big picture	2
2	Data and measurement: macro variables, yields, and monetary policy	2
2.1	Observed policy rate versus target rate	2
2.2	Sample and data sources	3
2.3	State variables and observables	3
2.4	Forward premiums	3
3	Models and empirical specifications	4
3.1	State dynamics	4
3.2	Nominal pricing kernel	4
3.3	Bond yields and forward premiums	4
3.4	Macro variables and the real pricing kernel	5
3.5	Taylor rule plus discretion	5
3.6	The identification problem	5
3.7	Long-run real output restriction	5
3.8	Long-run real asset-pricing restriction	6
3.9	Estimation sequence	6
3.10	Decomposing forward premiums	6
3.11	Dynamic correlations	7
4	Identification and empirical design	7
4.1	What identifies discretion	7
4.2	Why the short-rate equation is not enough	7
4.3	Why long-run restrictions are useful	7
4.4	Why asset pricing adds identifying power	8
4.5	What the paper can and cannot claim	8
5	Tables and figures	8
5.1	Figure 1 and Table 1: data and GMM estimation	8
5.2	Figure 2 and Figure 3: policy decomposition and the risk contribution of discretion	9
5.3	Figure 4 and Figure 5: dynamic correlations for macro variables and interest rates	10
5.4	Figure 6 and Figure 7: yield-curve factors and risk premiums	11
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** How can we separate the part of monetary policy that follows a rule from the part that reflects discretionary departures from that rule, and how does each component relate to macroeconomic and financial-market risk?
- **Core distinction.** The observed policy rate is decomposed into a target rate implied by a Taylor-style rule and a discretionary policy component:

$$i_t = i_t^R + s_t.$$

- **Identification problem.** The short-rate equation alone cannot identify the Taylor-rule parameters and discretion. The reason is that the same observed movement in the policy rate can be attributed either to the rule or to discretionary policy.
- **Main identification idea.** The paper imposes two long-run neutrality restrictions on discretion: a temporary discretionary monetary policy shock should not permanently affect real output and should not permanently affect real asset values.
- **Model setting.** The authors estimate an arbitrage-free macro term structure model with latent state variables, nominal yields, inflation, output growth, and a time-varying pricing kernel.
- **Why asset pricing matters.** The usual long-run output restriction gives only one restriction. The paper adds a long-run *real asset-pricing* restriction, using the permanent component of the real pricing kernel, to obtain the second restriction needed to separate rule and discretion.
- **Estimated Taylor rule.** The estimated inflation response is conventional and satisfies the Taylor principle: $\tau_\pi = 1.6230$. The estimated output-growth response is $\tau_g = 0.6532$.
- **Discretion is economically important.** Discretion contributes little to average short-maturity forward premiums, but contributes about 20% of the mean long-maturity forward premium and roughly 15–20% of the variance of forward premiums across maturities.
- **Macro result.** Both target-rate easing and discretionary easing coincide with good news about inflation. But for output growth, target-rate easing coincides with bad news, while discretionary easing coincides with good news.
- **Yield-curve result.** Target-rate shocks are related to the level of the yield curve. Discretionary-policy shocks are more closely related to the slope of the yield curve.
- **Risk-premium result.** Discretionary easing tends to lower short-term nominal bond risk premiums but raise long-term nominal bond risk premiums.
- **Important caveat.** The paper identifies a *process* for discretionary policy, not an exogenous monetary policy innovation. Therefore, the dynamic objects are correlations, not standard structural impulse responses.
- **Economic takeaway.** Monetary policy risk is not just about the observed short rate. The rule and the discretionary component have different implications for inflation, output growth, term premiums, and the yield curve.

2 Data and measurement: macro variables, yields, and monetary policy

2.1 Observed policy rate versus target rate

- The observed short rate is the policy rate, i_t .
- The target rate, i_t^R , is the part implied by a Taylor-style rule.

- The discretionary component, s_t , is the departure of the actual policy rate from the rule.

$$i_t = i_t^R + s_t.$$

Interpretation: The observed policy rate combines systematic responses to inflation and output growth with additional departures that the paper calls discretion. Ignoring this decomposition mixes two economically different sources of policy risk.

2.2 Sample and data sources

- The empirical sample runs from 1980Q3 to 2019Q4.
- The starting date is chosen to begin after the Volcker regime had begun to establish credibility.
- The short rate comes from Fama–Bliss data, available through CRSP.
- Longer-maturity zero-coupon Treasury yields come from Gürkaynak, Sack, and Wright.
- Real GDP growth comes from the National Income and Product Accounts.
- Core CPI inflation comes from the Bureau of Labor Statistics.
- Growth and inflation are measured as continuously compounded annual rates.

Interpretation: The model is estimated on macroeconomic variables and the nominal term structure jointly. That joint estimation is essential because asset-pricing restrictions are used to identify monetary policy discretion.

2.3 State variables and observables

The model uses an n -dimensional state vector x_t . The observed variables are linear functions of that state.

- Short rate:

$$i_t = a_0 + a^\top x_t.$$

- Inflation:

$$\pi_t = b_0 + b^\top x_t.$$

- Real output growth:

$$g_t = c_0 + c^\top x_t.$$

- Yields and forward rates are also affine in x_t .

Interpretation: The latent state vector is not given a direct economic name at the outset. Economic interpretation arrives only after the monetary policy rule and the long-run restrictions are imposed.

2.4 Forward premiums

Let $f_t^{(h)}$ be the forward rate h periods ahead and let i_{t+h} be the future short rate. The forward premium is

$$\text{fp}_t^{(h)} = f_t^{(h)} - \mathbb{E}_t i_{t+h}.$$

Interpretation: The forward premium is the model’s measure of nominal bond risk compensation. It is where the paper connects monetary policy discretion to asset pricing.

3 Models and empirical specifications

3.1 State dynamics

The state evolves according to

$$x_{t+1} = Ax_t + \varepsilon_{t+1}, \quad (1)$$

where

$$\varepsilon_t \sim N(0, \Sigma), \quad \varepsilon_t = Bw_t, \quad w_t \sim \text{iid}N(0, I).$$

Interpretation: The matrix A governs physical persistence and B governs the covariance structure of innovations. The paper does not assume these innovations are orthogonal in economically meaningful ways.

3.2 Nominal pricing kernel

The log nominal pricing kernel is

$$-m_{t+1}^{\$} = a_0 + a^\top x_t + \frac{1}{2} \lambda_t^\top \lambda_t + \lambda_t^\top w_{t+1}, \quad (2)$$

with

$$\lambda_t = \lambda_0 + \lambda x_t.$$

The one-period nominal interest rate is therefore

$$i_t = -\log \mathbb{E}_t \exp(m_{t+1}^{\$}) = a_0 + a^\top x_t. \quad (3)$$

Interpretation: The price of risk is time-varying. This is why asset-pricing restrictions can add information that would be absent in a constant-risk-premium model.

3.3 Bond yields and forward premiums

Absence of arbitrage implies that the price of a default-free zero-coupon bond is affine in the state:

$$-\log q_t^{(h)} = B_0^{(h)} + B^{(h)\top} x_t. \quad (4)$$

The yield is

$$y_t^{(h)} = \frac{-\log q_t^{(h)}}{h}. \quad (5)$$

The forward premium is affine in x_t :

$$\text{fp}_t^{(h)} = a^\top A^{*(h)} A_0^* - \frac{1}{2} a^\top A^{*(h)} B B^\top A^{*(h)\top} a + a^\top (A^{*h} - A^h) x_t. \quad (6)$$

Here

$$A^* = A - B\lambda, \quad A_0^* = -B\lambda_0,$$

are risk-neutral state dynamics.

Interpretation: The forward premium depends on the wedge between physical dynamics A and risk-neutral dynamics A^* . This wedge is the asset-pricing content of the model.

3.4 Macro variables and the real pricing kernel

Inflation and output growth are linear in the state:

$$\pi_t = b_0 + b^\top x_t, \quad g_t = c_0 + c^\top x_t. \quad (7)$$

The real pricing kernel is obtained by subtracting inflation from the nominal pricing kernel:

$$-m_{t+1} = -m_{t+1}^\$ - \pi_{t+1}. \quad (8)$$

Expanded in the paper's notation,

$$-m_{t+1} = (a_0 - b_0) + (a^\top - b^\top A)x_t + \frac{1}{2}\lambda_t^\top \lambda_t + (\lambda_t^\top - b^\top B)w_{t+1}. \quad (9)$$

Interpretation: Even without observing real asset prices directly, the real pricing kernel imposes restrictions on how policy discretion can affect long-run real valuations.

3.5 Taylor rule plus discretion

The target rate is modeled as

$$i_t^R = \tau_0 + \tau_\pi \pi_t + \tau_g g_t. \quad (10)$$

The observed policy rate is

$$i_t = i_t^R + s_t, \quad (11)$$

where discretion is

$$s_t = d^\top x_t. \quad (12)$$

Interpretation: Discretion can be persistent, transitory, tied to all state variables, or tied to only some state variables. The paper intentionally does not model its source structurally.

3.6 The identification problem

Substituting the linear observables into the policy equation gives

$$a = \tau_\pi b + \tau_g c + d, \quad (13)$$

$$a_0 = \tau_0 + \tau_\pi b_0 + \tau_g c_0. \quad (14)$$

Interpretation: The policy equation adds $n + 1$ restrictions but also adds $n + 3$ unknowns: τ_0 , τ_π , τ_g , and d . Two additional restrictions are needed to identify the Taylor-rule coefficients and the discretion process.

3.7 Long-run real output restriction

The first restriction says a temporary shock to discretion should not permanently affect the level of real output. Because output growth is stationary,

$$\log Y_{t+n} = \log Y_t + g_{t+1} + \cdots + g_{t+n}.$$

The long-run covariance between output and the discretionary shock is

$$\lim_{n \rightarrow \infty} \text{Cov}_t(\log Y_{t+n}, s_{t+1}) = c^\top (I - A)^{-1} B B^\top d.$$

The identifying restriction is

$$c^\top (I - A)^{-1} B B^\top (a - \tau_\pi b - \tau_g c) = 0. \quad (15)$$

Interpretation: This is a standard long-run neutrality restriction: discretionary policy can affect output growth in the short run, but it cannot permanently change the level of real output.

3.8 Long-run real asset-pricing restriction

The second restriction says a temporary discretionary shock should not permanently affect real asset values. The paper decomposes the real pricing kernel into permanent and transitory components,

$$m_{t+1} = m_{t+1}^P + m_{t+1}^T,$$

then requires the discretionary shock to be uncorrelated with the permanent component:

$$\mathbb{E}_t[m_{t+1}^P s_{t+1}] = 0. \quad (16)$$

At the long-run mean of the state, this becomes

$$[(b - a)^\top (I - A^*)^{-1} - \lambda_0^\top B^{-1}] BB^\top (a - \tau_\pi b - \tau_g c) = 0. \quad (17)$$

Interpretation: This is the paper's key innovation. It uses long-run restrictions from asset pricing, rather than only quantities, to identify policy discretion.

3.9 Estimation sequence

The empirical procedure is sequential.

1. Estimate the macro term structure model using GMM and quarterly data on yields, inflation, and output growth.
2. Treat the estimated macro term structure parameters as inputs into the two long-run restrictions.
3. Solve the two restrictions for τ_π and τ_g .
4. Recover

$$\tau_0 = a_0 - \tau_\pi b_0 - \tau_g c_0, \quad d = a - \tau_\pi b - \tau_g c.$$

Interpretation: The term structure model is estimated first; the monetary policy interpretation is imposed only afterward through the two long-run restrictions.

3.10 Decomposing forward premiums

Let

$$a^R = \tau_\pi b + \tau_g c$$

be the target-rate loading. Since $a = a^R + d$, the forward premium can be decomposed into a target-rule component, a discretion component, and an interaction term.

For the unconditional mean:

$$\mathbb{E}[\text{fp}_t^{(h)}] = (a^R)^\top A^{*(h)} A_0^* - \frac{1}{2} (a^R)^\top A^{*(h)} BB^\top A^{*(h)\top} a^R \quad (18)$$

$$+ d^\top A^{*(h)} A_0^* - \frac{1}{2} d^\top A^{*(h)} BB^\top A^{*(h)\top} d \quad (19)$$

$$- (a^R)^\top A^{*(h)} BB^\top A^{*(h)\top} d. \quad (20)$$

For the variance:

$$\text{Var}[\text{fp}_t^{(h)}] = (a^R)^\top (A^{*h} - A^h) V_x (A^{*h} - A^h)^\top a^R \quad (21)$$

$$+ d^\top (A^{*h} - A^h) V_x (A^{*h} - A^h)^\top d \quad (22)$$

$$+ 2(a^R)^\top (A^{*h} - A^h) V_x (A^{*h} - A^h)^\top d. \quad (23)$$

Interpretation: This decomposition measures how much of nominal term-premium levels and volatility are associated with systematic policy versus discretion.

3.11 Dynamic correlations

For any variable

$$z_t = \beta^\top x_t,$$

the dynamic covariance with a policy-rate shock is

$$\text{Cov}_t(z_{t+j}, i_{t+1} - \mathbb{E}_t i_{t+1}) = \beta^\top A^{j-1} B B^\top a. \quad (24)$$

The corresponding covariance with the target-rate shock is

$$\text{Cov}_t(z_{t+j}, i_{t+1}^R - \mathbb{E}_t i_{t+1}^R) = \beta^\top A^{j-1} B B^\top a^R. \quad (25)$$

The corresponding covariance with discretionary policy is

$$\text{Cov}_t(z_{t+j}, s_{t+1} - \mathbb{E}_t s_{t+1}) = \beta^\top A^{j-1} B B^\top d. \quad (26)$$

Interpretation: These are not impulse responses. They are dynamic correlations implied by the estimated macro term structure model.

4 Identification and empirical design

4.1 What identifies discretion

- The term structure model identifies the joint dynamics of the state vector, inflation, output growth, yields, and risk prices.
- The Taylor rule expresses the target rate as a function of inflation and output growth.
- The residual part of the short rate is discretion, but it cannot be measured without first identifying the Taylor-rule coefficients.
- The two long-run restrictions supply exactly the additional restrictions needed to estimate the rule and discretion.

Interpretation: The paper's identification does not come from a narrative shock, high-frequency monetary surprise, or exclusion restriction. It comes from long-run neutrality in quantities and real asset values.

4.2 Why the short-rate equation is not enough

- A rise in the policy rate when inflation rises could reflect a Taylor-rule response.
- It could also reflect a discretionary tightening that changes inflation endogenously.
- Since inflation, output growth, yields, and risk premiums are jointly determined, the short-rate equation alone does not separate rule from discretion.

Interpretation: This is the same logic behind the broader Cochrane critique: Taylor-rule parameters are not automatically identified just because a short-rate regression can be estimated.

4.3 Why long-run restrictions are useful

- They allow discretion to have short-run macroeconomic effects.
- They rule out permanent real effects of temporary discretionary policy shocks.
- They are broadly consistent with many New Keynesian and neoclassical models.
- They avoid imposing a fully structural theory of why discretion occurs.

Interpretation: The restrictions are deliberately minimal. They are strong enough to identify the policy decomposition but weak enough to preserve many possible theories of discretion.

4.4 Why asset pricing adds identifying power

- The output restriction only gives one restriction.
- The asset-pricing restriction focuses on the permanent component of the real pricing kernel.
- Since real asset values depend on the real pricing kernel, discretion cannot permanently move real asset values if it is long-run neutral.
- This restriction depends on risk-price parameters, so a time-varying pricing-kernel model is essential.

Interpretation: Monetary policy identification is usually treated as a macro question. This paper shows that asset pricing can help identify monetary policy because real risk prices embed long-run restrictions.

4.5 What the paper can and cannot claim

- The paper identifies a process s_t for discretionary policy.
- It does not identify a shock that is exogenous and orthogonal to macro and financial shocks.
- Therefore, the figures report correlations, not causal impulse responses.
- The findings are still informative because they reveal how the discretionary component of policy co-moves with inflation, output growth, yields, and risk premiums.

Interpretation: The right reading is: *conditional on the paper’s long-run restrictions, the data imply these relationships between rule-based policy, discretion, and financial-market conditions.*

5 Tables and figures

This section preserves the “Read” plus “Economic message” style. Adjacent visuals are grouped where possible to save space and speed comparison. All visuals are directly cropped from the original paper.

5.1 Figure 1 and Table 1: data and GMM estimation

Read: Figure 1:

- The upper panel plots U.S. real GDP growth and CPI inflation from 1980Q3 to 2019Q4.
- GDP growth is volatile around recessions, while inflation falls from the high early-1980s level to a lower and more stable post-Volcker range.
- The lower panel plots the short rate and longer zero-coupon yields.
- Yields decline substantially over the sample, while recessions and the zero lower bound show up clearly in the short rate.

Read: Table 1:

- The table reports GMM estimates of state dynamics, risk-neutral dynamics, macro loadings, and Taylor-rule parameters.
- Risk-neutral persistence is high: the first diagonal element of A^* is 0.9961.
- The estimated Taylor-rule inflation coefficient is $\tau_\pi = 1.6230$, satisfying the Taylor principle.

- The estimated output-growth coefficient is $\tau_g = 0.6532$.
- The estimated discretion loadings are $d^\top = (0.1868, 0.3783, 0.3973, 0.0177)$.

Economic message: The data show the macro-finance objects the model must jointly fit. The table shows that a conventional-looking Taylor rule can be recovered only after using the paper's long-run output and asset-pricing restrictions.

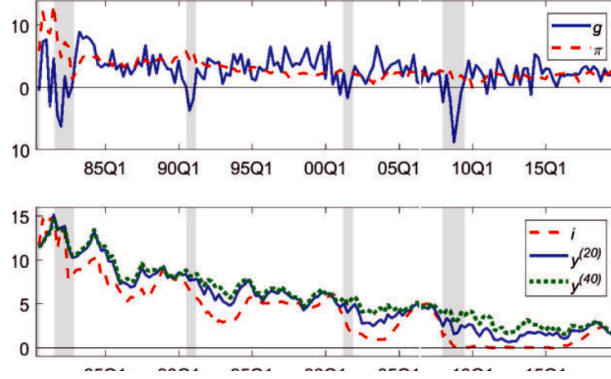


Figure 1: Figure 1: U.S. GDP growth, CPI inflation, and yields

Table 1
GMM estimation

A. State dynamics									
A					B				
0.9169 (0.0400)	0.2825 (0.0459)	0.2617 (0.0588)	0.6601 (0.1311)		0.0022 (0.0002)	0	0	0	
0.4680 (0.1885)	0.5501 (0.0623)	-0.0713 (0.0354)	-0.9401 (0.4700)		-0.0056 (0.0030)	0.0110 (0.0048)	0	0	
-0.4394 (0.1676)	0.1510 (0.0766)	0.7531 (0.0686)	0.6428 (0.4458)		0.0042 (0.0031)	-0.0011 (0.0051)	0.0022 (0.0002)	0	
0.0028 (0.0036)	-0.0668 (0.0215)	-0.0570 (0.0287)	0.4147 (0.0722)		-0.0004 (0.0003)	0.0010 (0.0005)	-0.0010 (0.0003)	0.0014 (0.0001)	
B. Term-structure model									
A*					λ_0	b	c		
0.9961 (0.0006)	0	0	0		-0.0625 (0.0053)	0.4947 (0.0060)	0.0158 (0.0071)		
0	0.8814 (0.0020)	0	0		-0.1421 (0.0191)	0.4243 (0.0126)	-0.1025 (0.0379)		
0	0	0.8348 (0.0035)	0		-0.0128 (0.0355)	0.4751 (0.0169)	-0.2578 (0.0438)		
0	0	0	0.3638 (0.0102)		-0.5803 (0.0777)	0.5927 (0.0181)	0.0311 (0.1221)		
C. Taylor rule									
τ_π	τ_g	$d^\top$							
1.6230 (0.3122)	0.6532 (0.2164)	0.1868 (0.1549)	0.3783 (0.1471)	0.3973 (0.1939)	0.0177 (0.1732)				

Based on moment restrictions (see Equations (3) and (4) in the Internet Appendix) for a sample period 1980Q3 to 2019Q4. Asymptotic standard errors are in parentheses. State dynamics: $x_{t+1} = Ax_t + Bw_{t+1}$. Macro term-structure model: $i_t = a_0 + a^\top x_t$, $\pi_t = b_0 + b^\top x_t$, $g_t = c_0 + c^\top x_t$, $h_t^{(h)} = B_0^{(h)} + B^{(h)} x_t$, and $B^{(h)} = a^\top (I - A^*)^{-1} (I - A^{*h})$, $B_0^{(h)} = a_0 + B_0^{(h-1)} - B^{(h-1)} B \lambda_0 - B^{(h-1)} B B^\top B^{(h-1)\top} / 2$. Policy: $i_t = \tau_0 + \tau_\pi \pi_t + \tau_g g_t + d^\top x_t$. The state

Table 1: Table 1: GMM estimation

5.2 Figure 2 and Figure 3: policy decomposition and the risk contribution of discretion

Read: Figure 2:

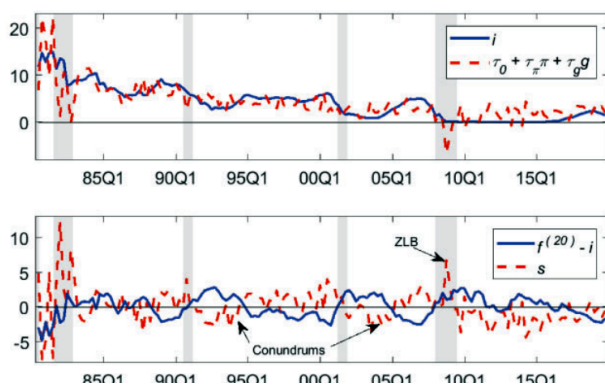
- The top panel compares the observed policy rate with the target rate implied by the estimated Taylor rule.
- The gap is discretionary policy, s_t .
- Discretion smooths the policy rate relative to the target rate.

- The figure shows large discretionary tightening before recessions and discretionary easing after recessions.
- The lower panel compares s_t with the five-year forward spread; discretion is strongly connected to yield-curve slope movements.

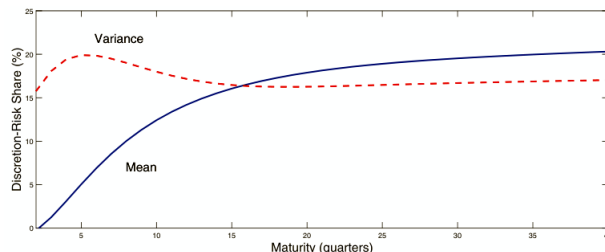
Read: Figure 3:

- Discretion contributes almost nothing to the average forward premium at short maturities.
- Its contribution rises with maturity and settles near 20% for long-maturity forward premiums.
- Discretion contributes more than 15% of forward-premium variance at all maturities and about 20% around one-year maturities.

Economic message: Discretion is not just a small residual in the short-rate rule. It is a quantitatively meaningful source of long-maturity term-premium risk.



(a) Figure 2 Target rate and discretionary policy shocks



(b) Figure 3 Forward premium share of discretionary policy

5.3 Figure 4 and Figure 5: dynamic correlations for macro variables and interest rates

Read: Figure 4:

- A policy-rate shock is positively correlated with inflation for many quarters.
- The output-growth correlation is muted in the aggregate because target-rate and discretionary components offset each other.
- Target-rate shocks are positively correlated with output growth shocks.
- Discretionary-policy shocks are negatively correlated with output growth shocks.

Read: Figure 5:

- Policy-rate shocks are persistently correlated with future short rates.
- The five-year yield is positively and persistently correlated with shocks to the policy rate.
- The long-yield correlation is driven by target-rate shocks, not discretionary shocks.
- Discretion is much more important for the short end than the long end of the yield curve.

Economic message: Rule-based policy and discretionary policy are not interchangeable. They have different correlations with growth and with long-term yields.

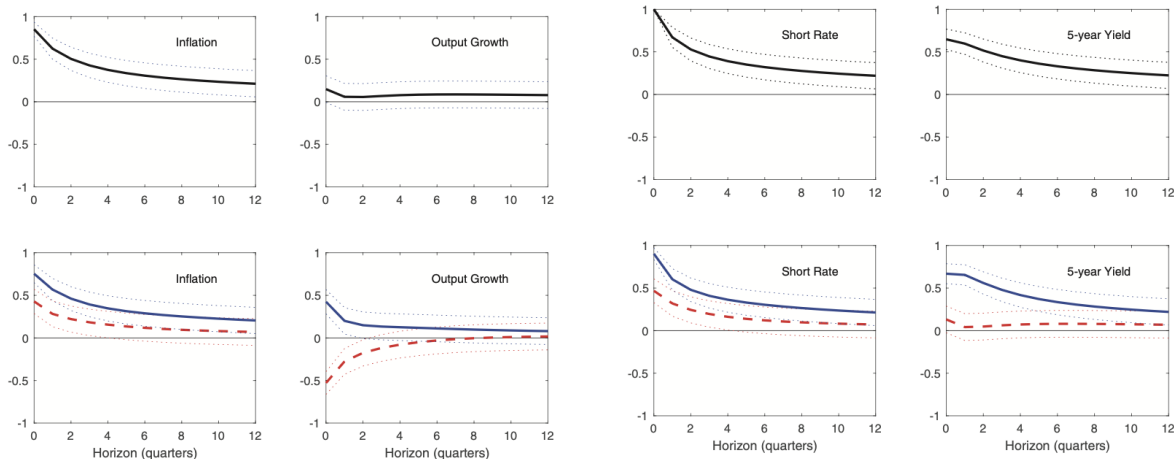


Figure 4
Dynamic correlations: Macro variables
The top panels plot the dynamic correlation with a shock to the policy rate for inflation and real output growth. The lower panels decompose this into the two sources for a policy shock, with the solid line representing a shock to the target rate and the dashed line representing a shock to policy discretion. Dotted lines represent 2-standard-deviation confidence bounds.

Figure 5
Dynamic correlations: Interest rates
The top panels plots the dynamic correlation with a shock to the policy rate for the short rate and the 5-year bond yield. The lower panels decompose this into the two sources for a policy shock, with the solid line representing a shock to the target rate and the dashed line representing a shock to policy discretion. Dotted lines represent 2-standard-deviation confidence bounds.

(a) Figure 4 Dynamic correlations: Macro variables

(b) Figure 5 Dynamic correlations: Interest rates

5.4 Figure 6 and Figure 7: yield-curve factors and risk premiums

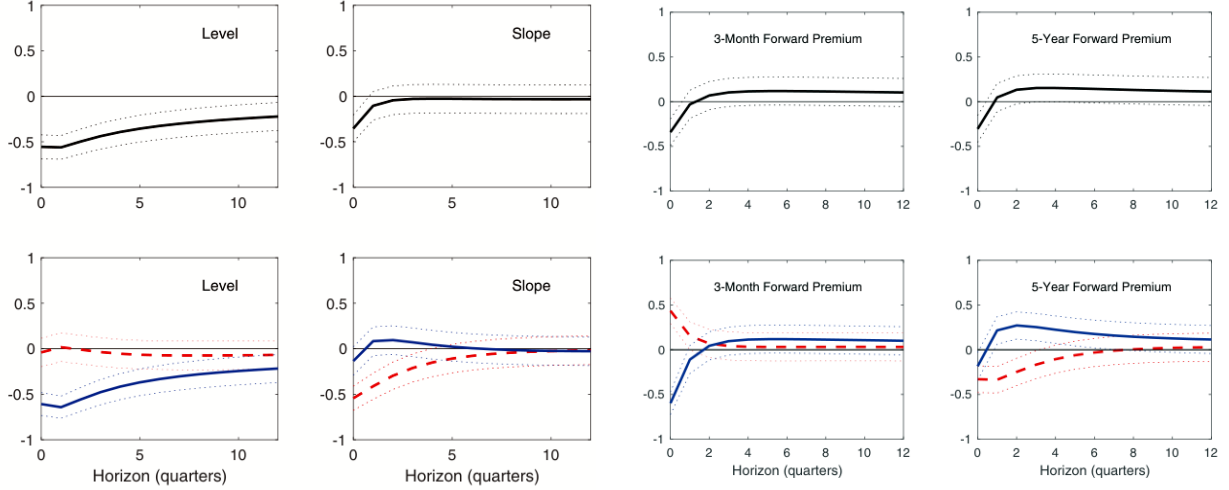
Read: Figure 6:

- Policy shocks are negatively correlated with the level factor in the figure's sign convention.
- The target-rate component is the main source of the level-factor relation.
- Discretionary policy is more closely tied to the slope factor.
- Discretionary easing tends to steepen the yield curve because it is more related to the short end than the long end.

Read: Figure 7:

- Aggregate policy-rate shocks hide offsetting effects from target and discretion.
- Target-rate easing is associated with bad news about financial conditions: higher risk premiums on both short- and long-maturity nominal bonds.
- Discretionary easing is associated with lower short-term risk premiums.
- Discretionary easing is also associated with higher long-term risk premiums.

Economic message: The paper's central financial result is visible here: shocks to the rule look like yield-curve level shocks, while shocks to discretion look more like slope and term-premium shocks.



(a) Figure 6 Dynamic correlations: Level and slope factors

(b) Figure 7 Dynamic correlations: Risk premiums

6 Short summary

- The paper studies monetary policy risk in a macro term structure model.
- The observed policy rate is decomposed into a Taylor-rule target rate and a discretionary component.
- The term structure model alone cannot identify the Taylor-rule coefficients and discretion.
- The paper uses two long-run restrictions: discretion has no permanent effect on real output and no permanent effect on real asset values.
- The second restriction uses the permanent component of the real pricing kernel, making asset pricing central to monetary policy identification.
- The estimated Taylor-rule parameters are conventional: $\tau_\pi = 1.6230$ and $\tau_g = 0.6532$.
- Discretion smooths the policy rate relative to the Taylor-rule target rate.
- Discretionary policy is an important contributor to term-premium risk, especially at longer maturities.
- Discretion accounts for roughly 15–20% of forward-premium variance.
- Target-rate and discretionary easing both coincide with good inflation news.
- Target-rate easing coincides with bad output-growth news, while discretionary easing coincides with good output-growth news.
- Target-rate shocks are related to yield-curve level movements.
- Discretionary-policy shocks are related to yield-curve slope movements.
- Discretionary easing tends to lower short-term nominal bond risk premiums but raise long-term nominal bond risk premiums.
- The paper does not identify exogenous monetary-policy shocks, so the reported dynamic patterns should be read as correlations rather than impulse responses.
- The main takeaway is that monetary policy discretion is a distinct source of macro-financial risk and should not be absorbed into a generic policy-rate shock.

7 Conclusion

7.1 Core conclusion

Backus, Chernov, Zin, and Zviadadze show that a macro term structure model with long-run real restrictions can separate systematic monetary policy from discretionary departures from that rule. The observed short rate is not enough to do this. The paper's asset-pricing restriction supplies the missing identifying information.

7.2 Economic intuition

A policy rate movement can come from two sources. The central bank may be following its systematic rule in response to inflation and output growth, or it may be departing from that rule. These two sources can produce similar movements in the short rate but very different relationships with output growth, long bond yields, and risk premiums. Asset prices help reveal the difference because they embed both expectations and risk compensation.

7.3 Takeaway

The paper's main lesson is that monetary policy risk should not be summarized by the policy rate alone. Rule-based policy and discretion have different macro and financial content. Target-rate shocks mostly look like yield-curve level movements. Discretionary shocks are linked more closely to slope movements and to changes in risk premiums, especially at long maturities.